



Comment on: Temperature-dependent conductivity in room temperature ionic liquids. The relative importance of diffusional and exchange mechanisms. [Simonis ED, Blanchard GJ (2026) *J Solid State Electrochem.* <https://doi.org/10.1007/s10008-025-06512-w>]

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Abstract

Simonis and Blanchard have reported tracer diffusion coefficients for the cationic chromophore rhodamine B in four ionic liquids (IL), using a Fluorescence Recovery After Photobleaching (FRAP) technique and employing a Stokesian hydrodynamic argument to deduce the self-diffusion coefficients of the constituent ions. The latter are used to calculate the Nernst-Einstein approximation to the conductivity which is compared to literature experimental data. Contrary to considerable experimental and computational evidence in the literature for the ILs examined, and statistical mechanics treatments of ion transport, they find that the conductivity exceeds the Nernst-Einstein estimate and suggest “exchange-mediated ion-transport” as a possible explanation for their observation. Examination of the data shows that their technique considerably underestimates the ion self-diffusion diffusion coefficients as conventionally measured by both steady and pulsed gradient spin-echo NMR and misestimates their temperature dependence. It is concluded that the FRAP technique and the Simonis-Blanchard treatment of the data give erroneous low estimates for ion self-diffusion in ionic liquids are the conclusion that additional mechanisms for charge transport beyond ionic vehicular motion is unwarranted.

Recently Simonis and Blanchard have reported in this Journal [1] that the (electrical) conductivity of room-temperature ionic liquids is not due solely to the motion of the constituent ions in an electric field but includes a contribution from an ion exchange mechanism involving ion association and ion pair dissociation that they call “ion-exchange mediated transport”. They derive ion self-diffusion coefficients of the ionic liquids from tracer diffusion coefficients for the chromophore cation rhodamine B measured by a *Fluorescence Recovery After Photobleaching* technique (FRAP) normally used for the determination of rotational correlation times. They then apply the Nernst-Einstein equation to determine an estimate of the conductivity of each ionic liquid over a range of temperature. As these values are much lower than literature conductivities, they conclude that an additional mechanism to conduction by vehicular ion movement in

an electric field—“exchange-mediated ion-transport”—is required to explain the difference.

That this contradicts extensive work in the literature based on both experiment and molecular dynamics simulations is the subject of the Comment. My criticism highlights that:

- The FRAP-derived ion self-diffusion coefficients are far too low compared to those obtained by spin-echo NMR. In the literature the Nernst-Einstein estimate of the conductivity always exceeds the experimental value due to the difference between velocity cross-correlations for cation–anion interactions and the mean of those for like-ion – like-ion interactions (cation-cation and anion-anion).
- The working equations are ill-defined, being based on simplistic hydrodynamics arguments, whereas the theory for spin-echo NMR is well established.
- The temperature dependence of the FRAP-derived ion self-diffusion coefficients is inconsistent with that of the molar conductivity. Normally for ionic liquids, Stokes-Einstein-Sutherland plots ($\log$ self-diffusion- $\log$

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fluidity) and Walden plots (log molar conductivity-log fluidity) have equal slopes.

- d) The concept of “exchange-mediated ion-transport” is not in accord with statistical mechanical and non-equilibrium thermodynamics treatments of ionic liquid and electrolyte transport properties.

These arguments are developed below.

The exchange contribution to the ionic liquid conductivity is obtained by subtraction of the Nernst-Einstein molar conductivity (Λ_{NE}) from experimentally determined molar conductivities (Λ) obtained from the literature. Λ_{NE} is given in terms of the cation and anion self-diffusion coefficients (D_{Si}) by

$$\Lambda_{\text{NE}} = \frac{F^2}{RT} (\nu_+ z_+^2 D_{\text{S}+} + \nu_- z_-^2 D_{\text{S}-}) \quad (1)$$

where ν_i and z_i are the ion stoichiometric and charge numbers, F is the Faraday, R the gas constant and T the temperature.¹ The detailed nature of the “exchange-mediated ion-transport” model is not made explicit, though it is argued that nanoscale structural heterogeneity or mesoscopic separation within the ionic liquid plays a part.

The self-diffusion coefficients reported by Simonis and Blanchard for several ionic liquids are derived from “fluorescence recovery after photobleaching” experiments with rhodamine-B, an ionic compound having chloride as the anion, as a chromophore tracer. Such experiments yield a symmetry averaged rotational correlational time and rotational diffusion coefficient of the chromophore ion at very low concentration in the solvent ionic liquid [2]. The rotational diffusion coefficient is used to estimate the chromophore tracer diffusion coefficient and the solvent ion self-diffusion coefficient using hydrodynamic (Stokes–Einstein) arguments.² So in this work, Λ is larger than Λ_{NE} .

This contradicts what has been reported in the literature, where Λ is always smaller than Λ_{NE} in both experiments [3–7] and molecular dynamics simulations [8–11]. It has long been known that the mobility of an ion in an electric field

differs from the diffusive mobility due to Brownian motion. The Nernst-Einstein (or Nernst-Hartley) equation is exact only for infinitely dilute electrolyte solutions. [14, 15] At higher concentrations one must take into account correlations between the ensemble and time averages of the velocities of the ions and the solvent molecules [16–18]. Where there is no solvent, as in neat ionic liquids, this reduces to three velocity cross-correlations, cation-cation, anion-anion and cation–anion, all of which are negative due to the conservation of momentum in the system, that is the ion ensemble and time averaged velocities are anti-correlated [11, 18]. This of course assumes that charge and mass are transported together (“vehicular motion”), [8–11, 16–18] that is, there is no charge transfer between cations and anions, something that has been confirmed experimentally. [19]

The ratio of the molar conductivity to the Nernst-Einstein limit (Y)³ is given by [5, 20]

$$\begin{aligned} Y &= \frac{\Lambda}{\Lambda_{\text{NE}}} \\ &= 1 + \frac{c(2\nu_+ \nu_- z_+ z_- f_{+-} + \nu_+^2 z_+^2 f_{++} + \nu_-^2 z_-^2 f_{--})}{(\nu_+ z_+^2 D_{\text{S}+} + \nu_- z_-^2 D_{\text{S}-})} \\ &= 1 - \frac{2c[f_{+-} - (f_{++} + f_{--})/2]}{(D_{\text{S}+}/\nu_+ + D_{\text{S}-}/\nu_-)} \end{aligned} \quad (2)$$

where c is the amount concentration or molarity. [The equality ($\nu_+ z_+ = -\nu_- z_-$) has been employed.] The velocity cross-correlation coefficients in the mass-fixed reference frame (VCC, f_{ij}) in these expressions were defined and expressed in terms of experimentally measurable quantities by Schönert [18] as:

$$f_{++} \equiv \frac{N_A V}{3} \int_0^\infty \langle v_{+\alpha}(0) \cdot v_{+\beta}(t) \rangle dt = \frac{RT\Lambda}{c} \left(\frac{M_-}{z_- FM} \right)^2 - \frac{D_{\text{S}+}}{\nu_+ c} \quad (3)$$

$$f_{--} \equiv \frac{N_A V}{3} \int_0^\infty \langle v_{-\alpha}(0) \cdot v_{-\beta}(t) \rangle dt = \frac{RT\Lambda}{c} \left(\frac{M_+}{z_+ FM} \right)^2 - \frac{D_{\text{S}-}}{\nu_- c}, \quad (4)$$

and

$$f_{+-} \equiv \frac{N_A V}{3} \int_0^\infty \langle v_{+\alpha}(0) \cdot v_{-\beta}(t) \rangle dt = \frac{RT\Lambda}{c} \frac{M_+ M_-}{z_+ z_- (FM)^2}, \quad (5)$$

where N_A is the Avogadro constant, V the volume of the ensemble, and M , M_+ , and M_- are the molar masses of salt, cation, and anion, respectively. The VCCs are time (t) integrals of the ensemble average of the velocity correlation functions of different ions as indicated by the subscripts α and β .

For the Simonis-Blanchard result to be true, the cation–anion velocity cross-correlation must be larger in magnitude than the mean value of the cation-cation and anion-anion

¹ The notation used here is slightly different to that employed by Simonis and Blanchard. It follows IUPAC recommendations and more usual literature usage.

² Tracer diffusion refers to the diffusion of a suitably labelled species (here the Rhodamine B cation) in an otherwise homogeneous environment (the ionic liquid). Self-diffusion refers to the diffusion of a suitably labelled species chemically indistinguishable from one of the species in the system e.g. a radiotracer atom may be the label. In NMR experiments the macroscopic magnetization vector resulting from the Boltzmann distribution of nuclear spins of an atomic species within the diffusing molecule or ion in an (applied) magnetic field acts as the label [12]. Depending on the mass of the labelling atom relative to that of the diffusing species and its position in the molecule there may be a mass or moment of inertia dependent “isotope effect” [13].

³ Sometimes called the inverse Haven ratio or ionicity [3, 4]

correlations, all being negative quantities. That this is not the case has been shown in numerous cases. [6, 7, 21–23]

The Simonis and Blanchard “exchange-mediated transport” is said to be analogous to the Grotthuss model often used to explain the anomalously high molar conductivities of the hydroxonium and hydroxide ions in H-bonded liquids and their solutions. [24] However, it has been shown that for the examples of water, formic acid and sulfuric acid, where the Grotthuss mechanism is generally accepted, and which can be formally treated as highly associated ionic liquids, the cation–anion VCC, f_{+-} , though very small is still negative. The Nernst-Einstein molar conductivity Λ_{NE} is anomalously high, but not the actual conductivities, $\kappa (= \Lambda c)$ and Λ , as the ion concentrations are extremely small [25]. Consequently, the ratio Y is close to zero. The pseudo-protic ionic liquid *N*-methylimidazolium acetate, a mixture of ionic and molecular species, with $Y \sim 0.06$, behaves similarly. [26]

Simonis and Blanchard argue that conduction can occur through the exchange of ions through dissociation of an ion pair followed by association of a given ion with a different neighbouring co-ion. However, there is no direct evidence for ion-pair formation in any of the aprotic ionic liquids examined. [19] If there were ion-pair formation, the Laity resistance coefficients (r_{ij}) for the cation-cation and anion-anion interactions would be negative. [6, 25, 27, 28] They are in fact positive. [6, 7] The relevant expressions, derived from nonequilibrium thermodynamics, are

$$r_{+-} = z_+ \nu_+ (z_+ + |z_-|) F^2 / \Lambda \quad (6)$$

for the cation–anion resistance coefficient and

$$r_{ii} = \frac{1}{|z_j|} \left[\frac{(z_+ + z_-)RT}{D_{Si}} - |z_i| r_{+-} \right], i = +, -; j \neq i \quad (7)$$

for the like ion coefficients. r_{+-} is necessarily positive, but the two r_{ii} may be positive or negative.

The Simonis-Blanchard thesis requires diffusion coefficients determined by their methods to be identical to self-diffusion coefficients determined by other means. This aspect is examined as follows.

The working equations for the FRAP method employed by Simonis and Blanchard are not well defined: they rely on an assumed hydrodynamic relationship between rotational and translational diffusion coefficients and estimates of the hydrodynamic radii for both the chromophore species and the constituent ions of the ionic liquid component. The appears to be no distinction made between the interactions of the rhodamine B cation with the ionic liquid cation and anion despite the difference in charge. A one-to-one coupling of rotational and translational diffusion is also assumed. The Stokes–Einstein equation employed in ref.

[1] and [2] strictly speaking applies to a solute at infinite dilution with a continuum of a certain viscosity. It is a poor model for a charged solute in a molten salt with strong Coulombic interactions.

On the other hand, the theory for measurements of self-diffusion by spin-echo NMR is well developed and rigorous. [29, 30] It should be noted that NMR methods have been checked against radio-tracer capillary and diaphragm cell methods in many systems, both electrolyte [31] and non-electrolyte, [29] so these have a high level of acceptance.

It is best practice that authors compare their measured data to published literature values where available. “If new data are inconsistent with previously published results, the difference should be discussed, as well as all concerns about the correctness of the new result”. [32] Unfortunately, this has not been done in ref. [1], though in an earlier paper [2] Simonis and Blanchard have asserted that the ion self-diffusion coefficients derived from their FRAP measurements for 1-methyl-3-butylimidazolium bis(trifluoromethylsulfonyl)amide, [BMIM] [(CF₃SO₂)₂N] or [BMIM][TFSI], (5.4 and $5.1 \cdot 10^{-12}$ m²s⁻¹ for the cation and anion, respectively, at 293 K) are consistent with literature pulsed gradient spin-echo NMR measurements [4]. This is not the case: the values they cite (16 and $11 \cdot 10^{-12}$ m²s⁻¹ at 303 K) are for a *different* ionic liquid, namely 1-methyl-3-butylimidazolium bis(pentafluoroethylsulfonyl)amide, [BMIM] [(C₂F₅SO₂)₂N]. The data in Table 2 in the Supporting Information of ref. [4] appear to have been misread. Measurements from this laboratory for [BMIM][TFSI] made by steady gradient spin-echo NMR (which has fewer experimental variables than pulsed gradient NMR) [5, 29] yield 22.0 and $16.7 \cdot 10^{-12}$ m²s⁻¹ for the cation and anion, respectively, at 293 K, consistent with the Tokuda measurements within the combined uncertainties. Simonis’ and Blanchard’s values are far too low, which is why their estimates of Λ_{NE} are lower than the experimental molar conductivity. Figures 1, 2, 3, and 4 show this for the four ionic liquids examined in the paper under consideration here. It is also clear that the FRAP measurements have a weaker temperature dependence than that of NMR spin-echo self-diffusion measurements, whereas the latter correlates exactly with that of the molar conductivity in each case, that is, the slope of a $\ln(D_{S+} + D_{S-})$ versus $\ln \Lambda$ is unity, within experimental error [5–7]. Equivalently, the slopes of Stokes-Einstein-Sutherland plots for the two ions and of the Walden plot for the molar conductivity are all equal [5–7].

It should be noted that in Table S2 of the Supporting Information for Ref. [1] single sets of values are given for $D_S([BMIM]^+)$ and $D_S([TFSI]^-)$, that is no distinction is made between the two [BMIM]⁺ salts for the cation on

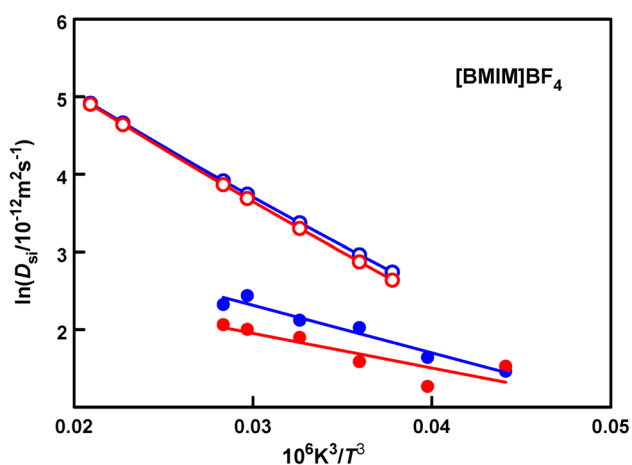


Fig. 1 FRAP-derived [1] and NMR spin-echo self-diffusion coefficients [7] for [BMIM]BF₄. Symbols: cation, blue; anion, red; closed, FRAP; open, NMR

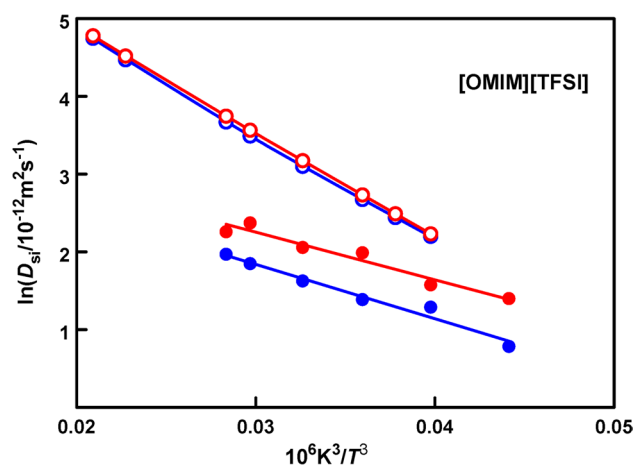


Fig. 4 FRAP-derived [1] and NMR spin-echo self-diffusion coefficients [6] for [OMIM][TFSI]. Symbols: as for Fig. 1

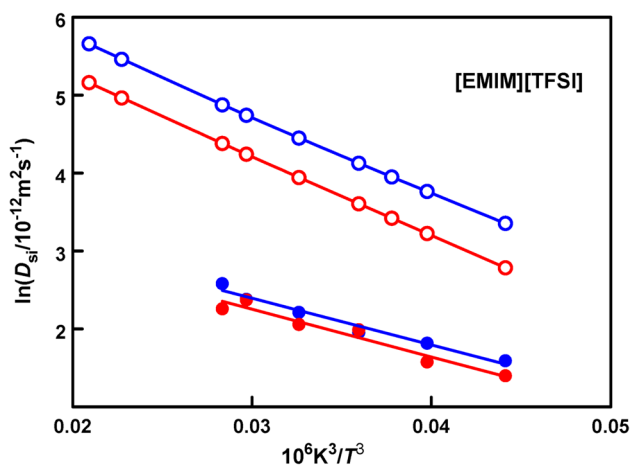


Fig. 2 FRAP-derived [1] and NMR spin-echo self-diffusion coefficients [6] for [EMIM][TFSI]. Symbols: as for Fig. 1

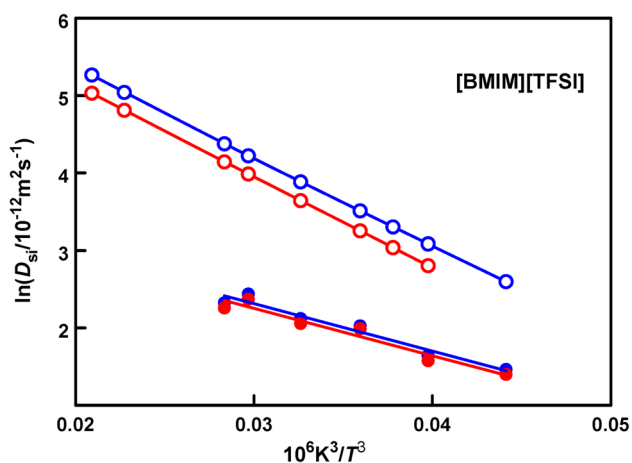


Fig. 3 FRAP-derived [1] and NMR spin-echo self-diffusion coefficients [6] for [BMIM][TFSI]. Symbols: as for Fig. 1

the one hand nor for the three [TFSI][−] salts on the other. Our work [6, 7] and that of the Watanabe group [3, 4] show clearly that the self-diffusion coefficient of a given ion in a series of ionic liquids varies with the nature of the co-ion. In this case, at 303 K for example, $D_S([BMIM]^+)$ is 19.4 and $33.4 \cdot 10^{-12} \text{ m}^2 \text{ s}^{-1}$ in [BMIM]BF₄ and [BMIM][TFSI] respectively and $D_S([TFSI]^-)$ is 36.8, 25.8 and $15.4 \cdot 10^{-12} \text{ m}^2 \text{ s}^{-1}$ in [EMIM][TFSI], [BMIM][TFSI] and [OMIM][TFSI] respectively. [6] The estimated relative standard uncertainty, including calibration of the magnetic field gradient coil, is 3%.

All in all, it seems that the FRAP measurements of the rotational relaxation time of a dilute chromophore in an ionic liquid cannot be used to estimate the ion self-diffusion coefficients. In consequence the conclusions of Simonis and Blachard that ion vehicular motion only partially contributes to the electrical conductivity and that additional mechanisms such as their “exchange-mediated ion-transport” model therefore need to be considered are unwarranted.

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